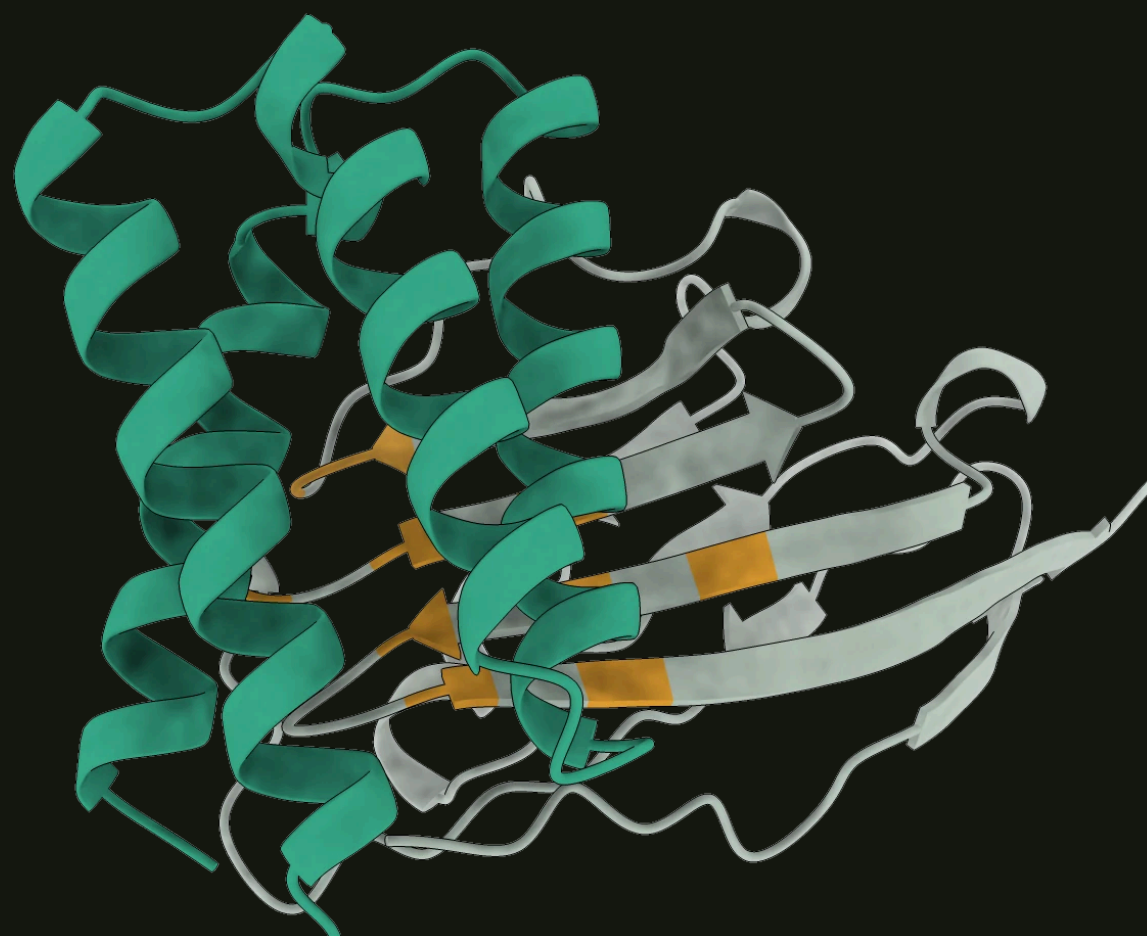


You already know your target.

“Design binders against PD-L1 to block its interaction with PD-1.”

No structure named, no epitope chosen. It ranked the nine solved structures of PD-L1 by their measured interfaces, proved the chain it picked was PD-L1 and not the binder crystallised against it, chose a ten-residue epitope, sized the campaign from its own trial, and returned **708 gated candidates** from 3,988 refolds — **\$0.18** of model spend and 21.4 GPU-hours, unattended.



Nine solved structures. Ranked on the interface, not the citation count.

ENTRY	Å	BSA Å ²	IFACE RES	H - BONDS	PHOB
7CZD anti-PD-L1 VHH	1.64	2,449	44	30	0.34
8AOK VHH 6	1.60	2,215	46	37	0.22
8ZNL PD-L1 de novo binder	1.77	1,844	31	17	0.39
5JDS Nanobody	1.70	1,648	30	23	0.30
8AOM VHH6	2.20	1,307	17	3	0.71

WHY NOT THE TWO ABOVE IT

7CZD — Alternative high-affinity VHH interface; retained as secondary competitive site in sites_json.

8AOK — VHH6 interface with lower hydrophobic fraction (phi = 0.22) compared to 8ZNL (phi = 0.39).

Every figure in that table is computed from coordinates before any model reads the list, so the choice is argued over measurements rather than over abstracts — 8ZNL, row 3: a validated de novo mini-binder on the PD-1-competitive face at 1.77 Å, hydrophobic fraction 0.39, taken over two entries that bury more surface. Burying the most surface is not the same as offering the most designable surface, and the rejections above are the stage’s own account of the trade.

Chain A of this structure is not the target. It is somebody else's binder.

8ZNL holds PD-L1 on chain B and a PD-L1 de novo binder on chain A. Told to design against “PD-L1”, an earlier version of this stage — on 7CZD, the VHH complex one row above this one — picked the partner chain and put its hotspots on the nanobody's own CDR loop.

100%

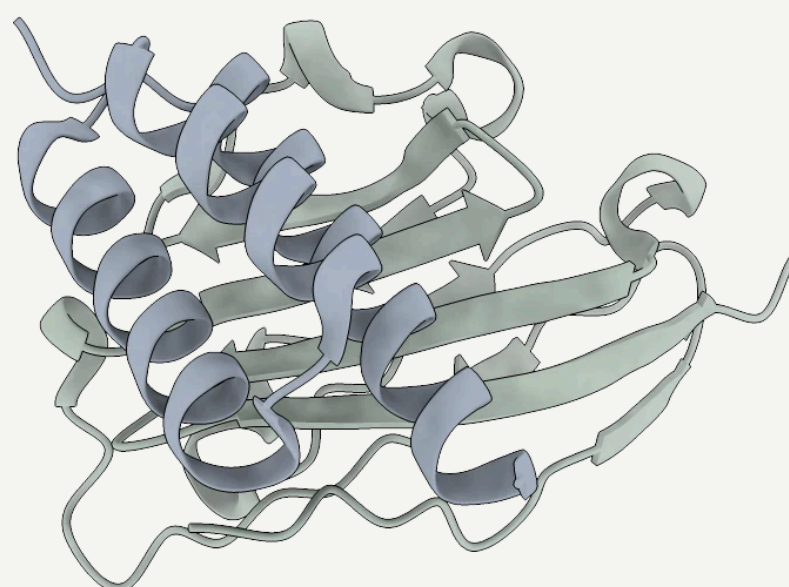
7CZD CHAIN B VS Q9NZQ7

20%

7CZD CHAIN A VS Q9NZQ7

WHAT RUNS NOW, BEFORE A TRIM IS BUILT

Every modelled residue of the chosen chain is aligned against the target's canonical UniProt sequence, and a mismatch halts the run. Checking the residues cannot catch this — they are real and correctly numbered, just on the wrong protein.



The crystallised partner bound to PD-L1 in 8ZNL, down the epitope axis — 1,844 Å² buried. Designing against its own surface instead of PD-L1's would have been geometrically unremarkable and biologically worthless.

Ten residues, on the face PD-1 itself binds.

10

HOTSPOTS DECLARED

2

CANDIDATE REGIONS
SCORED

excellent

80% HYDROPHOBIC

Ala53, Ile55, Tyr57, Val69, His70, Arg114, Met116, Ser118, Ala122, Tyr124 — the Central β -Sheet Hydrophobic & Aromatic Core, chosen over one alternative face. Tyr124 (-5.0), His70 (-3.7), Ser118 (-2.0) are the strongest of the 6 carrying a computed $\Delta\Delta G$. A region declares at most twelve: more is not stricter, because RFD3's hit rate falls as the set grows.

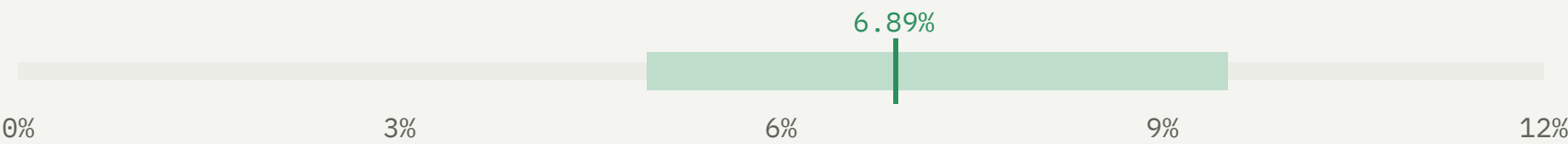


The epitope with no binder present. Residue names and sidechain atoms are read from the file at each position — a hotspot table can be internally consistent and still describe a different structure of the same protein.

The trim then did nothing, which is also a result: at 114 residues against a 220 budget the chain was kept whole, in 1 segment (70-86, /0, B20-133). Deciding a target already fits is as much that stage's job as cutting one down.

It measured its own hit rate before spending a day of GPU.

A 580-backbone trial, refolded 4 sequences deep: **33 of 479** backbones produced a design clearing every gate at the excellence bar.



95% Wilson interval 4.95–9.52%. At this rate a trial sees a handful of hits, and a point estimate off a handful is not a number worth four days — so the campaign is sized against the pessimistic end.

BASIS	BACKBONES	REFOLDS	GPU-HOURS
point estimate	879	2,903	10.0
Wilson lower bound	1,224	4,043	13.9

VERDICT · SCALE UP

It also raised its own success bar from 0.7 to **0.85**. An unusually good target should not be sized to the same fixed bar as a marginal one, so the bar climbs to the strictest rung the trial still supports and the budget still affords.

Eight gates, each applied to every refold on its own.

1,224

BACKBONES DIFFUSED

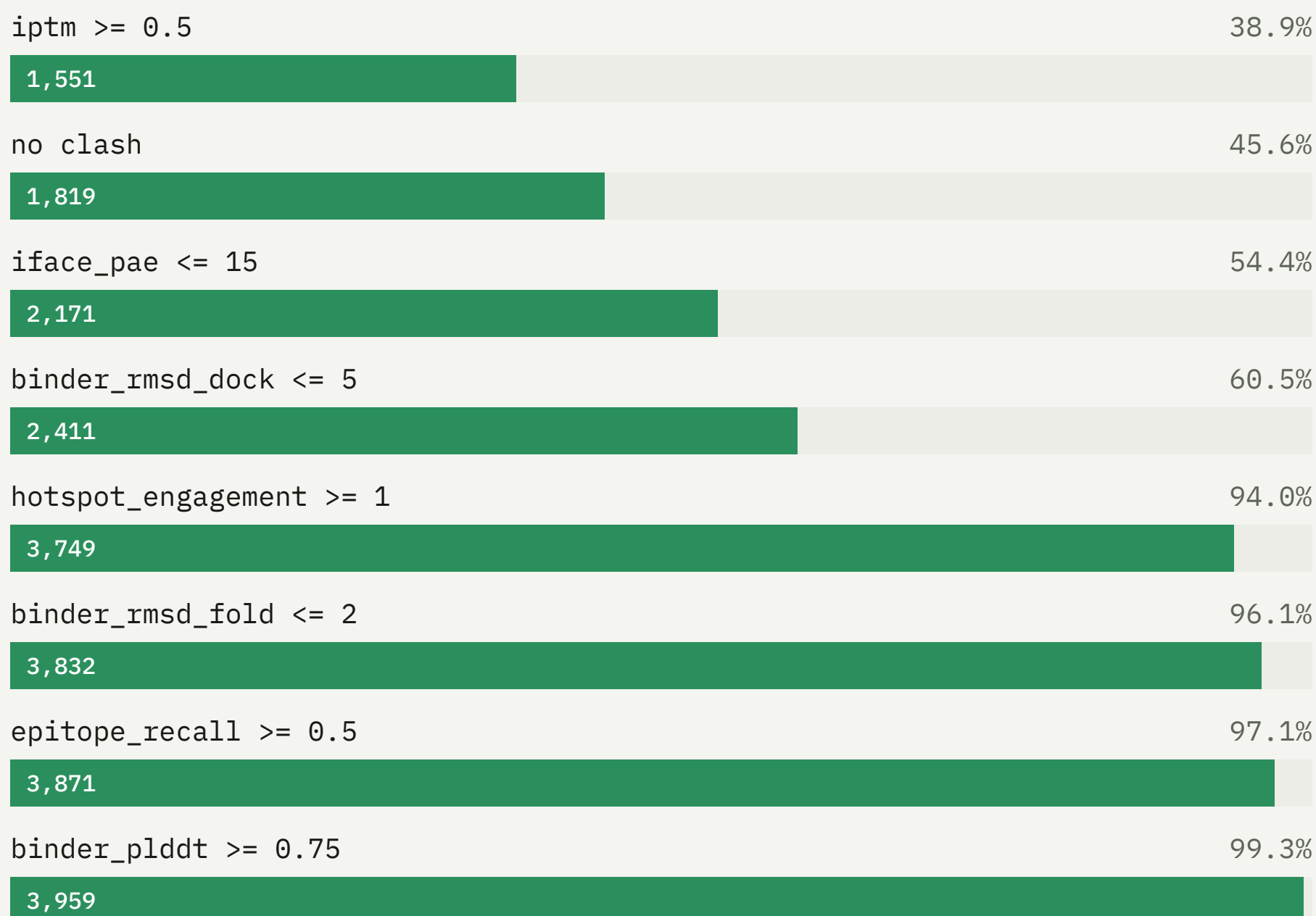
3,988

REFOLDS SCORED

708

SURVIVORS

A cheap sidecar prefilter kept 81% of the backbones (997), and each survivor was folded as 4 sequences. Bars are the share of all 3,988 refolds each gate would pass *alone*; the 708 survivors are the intersection, across 390 distinct backbones.



A model can be confident about an interface it invented.

The refolder is never shown the docked pose, so ipTM reports confidence in whatever interface the model chose — not in the one that was designed for. Rank on it alone and you select confidently mis-docked binders.

802

REFOLDS OVER IPTM 0.7

95%

OF THOSE DOCKED ≤ 5 Å

43

CONFIDENT AND MIS-DOCKED

THIS TARGET IS THE EASY CASE

95% is unusually good. Of designs over ipTM 0.7 on two other campaigns in this repo, only **45%** (8TAC) and **8%** (CD79b) were docked on target at all. The dock-RMSD gate is the one doing the work, and on a harder target it does nearly all of it.

The commonest single failure was geometric and it was rarely alone: 1,577 refolds failed dock-RMSD first, and 1,407 of those (89.2%) were below the ipTM gate as well.

It aimed at PD-1's own footprint, without ever being shown it.

Nothing in the campaign opened 4ZQK, the PD-1/PD-L1 complex. Superposed onto the top-ranked design afterwards (0.87 Å over 113 Cα, 100% identity), the two binders land on the same face.

12/15

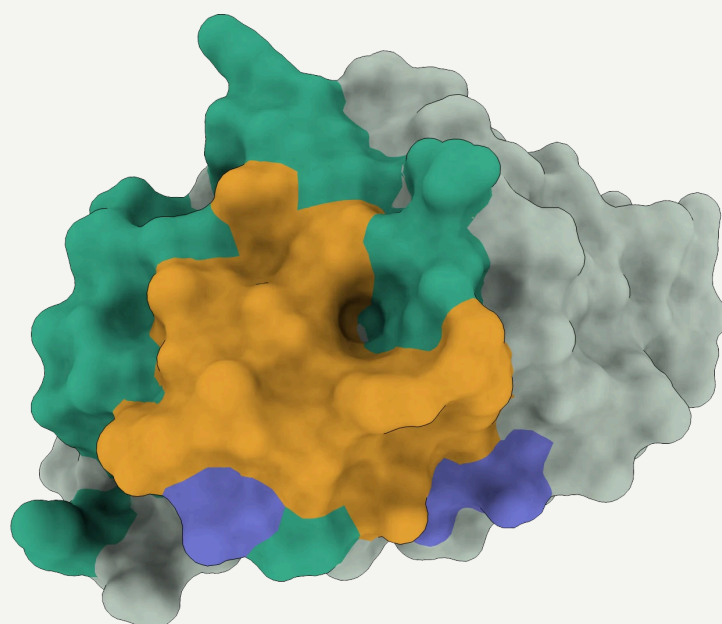
PD-1 CONTACTS ALSO
TOUCHED

80%

FOOTPRINT OVERLAP

6/10

HOTSPOTS ARE PD-1
CONTACTS



■ both (12) ■ PD-1 only (3) ■ design only (18) ■ neither

Computed at build time, not part of the run — the superposition and both footprints come out of the run's own rank-1 refold at a 4.5 Å heavy-atom cutoff, so they move as a reviewable diff like every other figure here. 4ZQK itself stays external to the campaign, which is what makes it a check.

RANK 1

86 residues, ipTM 0.861, dock-RMSD 1.23 Å, binder pLDDT 0.83, Rosetta $\Delta\Delta G$ -53 REU. One of 20 shortlisted, each from a different backbone.

THE BILL

Three reasoning stages. Three model calls. The rest is arithmetic.

\$0.18

MODEL SPEND, CAP \$10

21.4

GPU-HOURS, ONE CARD

21

HOURS WALL CLOCK

STAGE	MODEL CALLS	GPU-H
target intel — structures ranked, chain verified	1	—
interface — the epitope	1	—
trim, spec, pilot	0	1.10
calibration — the trial and its verdict	0	6.60
production, gating, ranking	0	13.67
summary — the write-up	1	—

ONE CALL PER STAGE, AND NO MODEL BUILT IN

An agentic stage bills once per turn that reaches the model, so three stages is not automatically three calls — an earlier run against this target billed the interface stage several times over, because the query it was handed never named the entry the orchestrator had *already downloaded* and the skill kept concluding no structure existed. The whole bill here is `gemini-3.7-flash`, the default since safety-classifier refusals on structural-biology prompts started costing real money for no output — and no model is built in, since Claude, Gemini and GPT each drive the same loop, tools and cost ledger.

Nothing here has been near a bench.

Every figure on this deck is computational: folding confidence, interface geometry, buried surface area, Rosetta terms over 300 relaxed poses. Designs like these need expression, purification and a binding assay before any of it is a result. The trial, the gates and the ranking exist to make that shortlist small and defensible — not to skip it.

And the code has moved since the run.

This campaign ran 2026-09-10–2026-09-11. One gate has been deliberately loosened since: `hotspot_engagement` went from 1.0 to 0.75, because requiring every hotspot rejected refolds for missing residues their own backbone never targeted. Re-gating the same 3,988 refolds through today’s code changes nothing here:

	SURVIVORS	BACKBONES	TOP 20
as it ran, 2026-09-11	708	390	—
re-gated today	708	390	identical

The loosened gate can only admit refolds, and it admitted none: all 4 refolds this campaign rejected on engagement fail another gate anyway. So the survivor set, its ranking and the top twenty are the same objects, and the figures above are the run’s own either way. On a campaign where the threshold mattered this slide would carry two different numbers — it is worth stating that it does not.

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Source-available under PolyForm
Noncommercial 1.0.0. Academic and
personal use, no fee, no negotiation.
Commercial use needs a separate licence.

Entry point shown here	<code>--workflow binder, target named</code>
The other one	<code>--workflow ppi, one sentence, no target</code>
And a third	<code>--workflow structure, your own file</code>
This campaign, written up in full	<code>campaign_pdl1.html</code>

Four campaigns documented end to end, every figure extracted from the
run that produced it — including the ones that make the tool look worse.